Streamlining Ticket Assignment for Efficient Support Operations (PDF-Based Inputs)

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1. Introduction

In many organizations, support tickets are received in **PDF format**, leading to inefficiencies such as manual data entry, delayed responses, and inconsistent ticket assignment.

This document outlines a streamlined approach to:

- Convert PDF-based tickets into structured data
- Automate classification and assignment
- Improve response time and operational efficiency

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2. Problem Statement

Handling tickets via PDFs results in:

- ❌ Manual extraction of information
- ❌ Increased processing time
- ❌ Higher chances of human error
- ❌ Inefficient workload distribution

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3. Objective

To design an automated system that:

- Extracts data from PDFs
- Classifies tickets intelligently
- Assigns tickets to appropriate teams
- Tracks performance metrics

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4. Proposed Solution

4.1 PDF Data Extraction

Convert PDF content into structured data using:

- OCR for scanned documents
- Parsing tools for digital PDFs

Key fields extracted:

- Ticket ID
- Customer Name
- Issue Description
- Category
- Priority
- Timestamp

Output format:

- JSON / CSV for easy processing

4.2 Ticket Classification

Rule-Based Approach

- Keyword matching (e.g., "payment" → Billing)
- Pattern recognition

AI-Based Approach

- Machine learning models for text classification
- Continuous improvement using historical data

4.3 Automated Ticket Assignment

Tickets are assigned based on:

- **Category:** Routed to relevant department
- **Priority:** High-priority tickets handled first
- **Workload:** Assigned to least busy agent
- **SLA Rules:** Ensures timely resolution

4.4 Workflow Automation

End-to-End Process:

1. PDF received via email/upload
2. Data extraction
3. Ticket classification
4. Auto-assignment
5. Notification to support agent

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5. System Architecture

Input Layer:

- Email / Upload Portal

Processing Layer:

- OCR + Data Parsing

Logic Layer:

- Classification Engine
- Assignment Engine

Output Layer:

- Ticketing System (Dashboard + Alerts)

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6. Benefits

- ✅ Reduced manual effort
- ✅ Faster ticket resolution
- ✅ Improved accuracy
- ✅ Better resource utilization
- ✅ Enhanced customer satisfaction

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7. Key Recommendations

- Standardize PDF formats for consistency
- Use templates instead of free-form documents
- Start with rule-based automation, then scale to AI
- Continuously monitor and optimize performance

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8. Conclusion

By converting PDF-based tickets into structured data and automating classification and assignment, organizations can significantly improve support efficiency, reduce response times, and deliver a better customer experience.